

Ubiquitous Computing and Edge Caching

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Ubiquitous Computing

1. **Edge Computing**
2. **Cloudlet**
3. **Fog computing**
4. **Internet of Things (IoT)**
5. **Virtual Conferencing**
6. **Virtual Events (3D and Hybrid)**

Ubiquitous computing

- Mark Weiser: Three basic ubiquitous computing devices:
 - Tabs: a wearable device that is approx in centimeters
 - Pads: a hand-held device that is approximately a decimeter in size
 - Boards: an interactive larger display device that is approximately a meter in size
- computing is made to appear anytime and everywhere
- any device, in any location, and in any format

Weiser, Mark. "The computer for the 21st century." *ACM SIGMOBILE mobile computing and communications review* 3.3 (1999): 3-11.

https://en.wikipedia.org/wiki/Ubiquitous_computing

Edge computing

- Distributed computing paradigm
- Computation and data storage closer to the user location
- Improve response times and save bandwidth
- Cloud computing operates on big data, whereas Edge computing operates on “instant data”
- Content Delivery Network or Content Distribution Network (CDN)
 - Akamai CDN
 - Akamai-Facebook’s Photo-Serving Stack

Cloudlet

- First coined by Mahadev Satyanarayanan (Satya), Victor Bahl, Ramón Cáceres, and Nigel Davie
- It is a mobility-enhanced small-scale cloud datacenter that is located at the edge of the Internet.
- It work as a *data center in a box* which *brings the cloud closer*.
- Support resource-intensive and interactive mobile applications by providing powerful computing resources to mobile devices with lower latency.

Fog computing

- Architecture that distributes computing, storage, control and networking functions closer to the users along a cloud-to-thing.
- Fog computing is often erroneously called edge computing, but there are key differences.
- Fog works with the cloud, whereas edge is defined by the exclusion of cloud.
- Fog is hierarchical where edge tends to be limited to a small number of layers.
- Cloud computing deal with Big Data, whereas Fog computing deals with real-time data generated by sensors or users.

¹ IEEE Standard Association. "IEEE 1934-2018-IEEE Standard for adoption of OpenFog reference architecture for fog computing." (2018).

https://en.wikipedia.org/wiki/Fog_computing

Internet of Things (IoT)

- The network of physical objects —“things”— embedded with sensors, software, and other technologies for the purpose of connecting and exchanging data with other devices and systems over the Internet.
- Example:
 - “Smart Home” devices and appliances,
 - “Smart city” equipment and facilities,
- Real-Time Data Analytics
- *Information explosion or Data Deluge*
 - due to *data flood* or *information flood*
 - ever-increasing amount of electronic data exchanged per time unit
 - unmanageable amounts of data growth V/S power of data processing

Virtual Conferencing

- Teleconference: Phone lines, Landlines or Cellular devices
- Video conference:
 - Webcam, Microphone, Speaker, Internet
 - hardware, software, devices is dedicated for this
- Web Conference: Internet and Cloud supported
 - Web 2.0
 - Well-Known: Google Meet, Skype, and Microsoft Team
 - Multi-Communications from Many sender to Many receivers
 - Webinars ("web seminars"), Webcasts (live media presentation), Podcast (audio presentation), and web meetings
 - **Virtual Events**

Virtual Events

- An online event involves people interacting in a virtual environment on the web, instead of physical meeting.
- Multi-session online events often feature webinars and webcasts.
- Aim to create similar experience as physical meeting.
- Live-streaming the event online or on-demand video.
- Issues
 - Echo of voice
 - Audio and Videography logistics
 - Network Bandwidth of conferencing server and users

2D Virtual Event: Live Streaming

The screenshot displays the Virtual Heidelberg Laureate Forum (HLF) 2020 interface. On the left is a navigation menu with options: Home, News (4), Stage/Livestream, My Sessions, Poster Gallery, Exhibition | I AM A.I., Film | Secrets of the Surface, and Exhibition | Remember M... The main content area is titled 'Detail' and shows a 'Stage' event: 'Dialogue: Hoare/Lamport' scheduled for 09/21/2020 from 6:30 PM to 7:20 PM. A note indicates the session begins at 10:00 PM in the user's time zone. Below the event details is a video player showing a live stream of two speakers, Tony Hoare and Leslie Lamport, in a split-screen format. The video player has a caption 'as my career goal.' and a subtitle 'Scientific Dialogue Sir C. Antony R. Hoare Leslie Lamport'. To the right of the video player is a list of speakers: Tony Hoare and Leslie Lamport. Below the video player is a comment section with a text input field, a 'Send' button, and a character count '1000 characters left'. It shows '2 Comments' and a sorting option 'chronologically'. The first comment is from Roy Levin, posted '3 minutes ago', mentioning that ACM recently published a substantial retrospective on Leslie Lamport's career and technical work as 'Concurrency: The Works'.

[Virtual HLF 2020 - Heidelberg Laureate Forum](#)

3D Virtual Event: Live Video



3D Virtual Event:



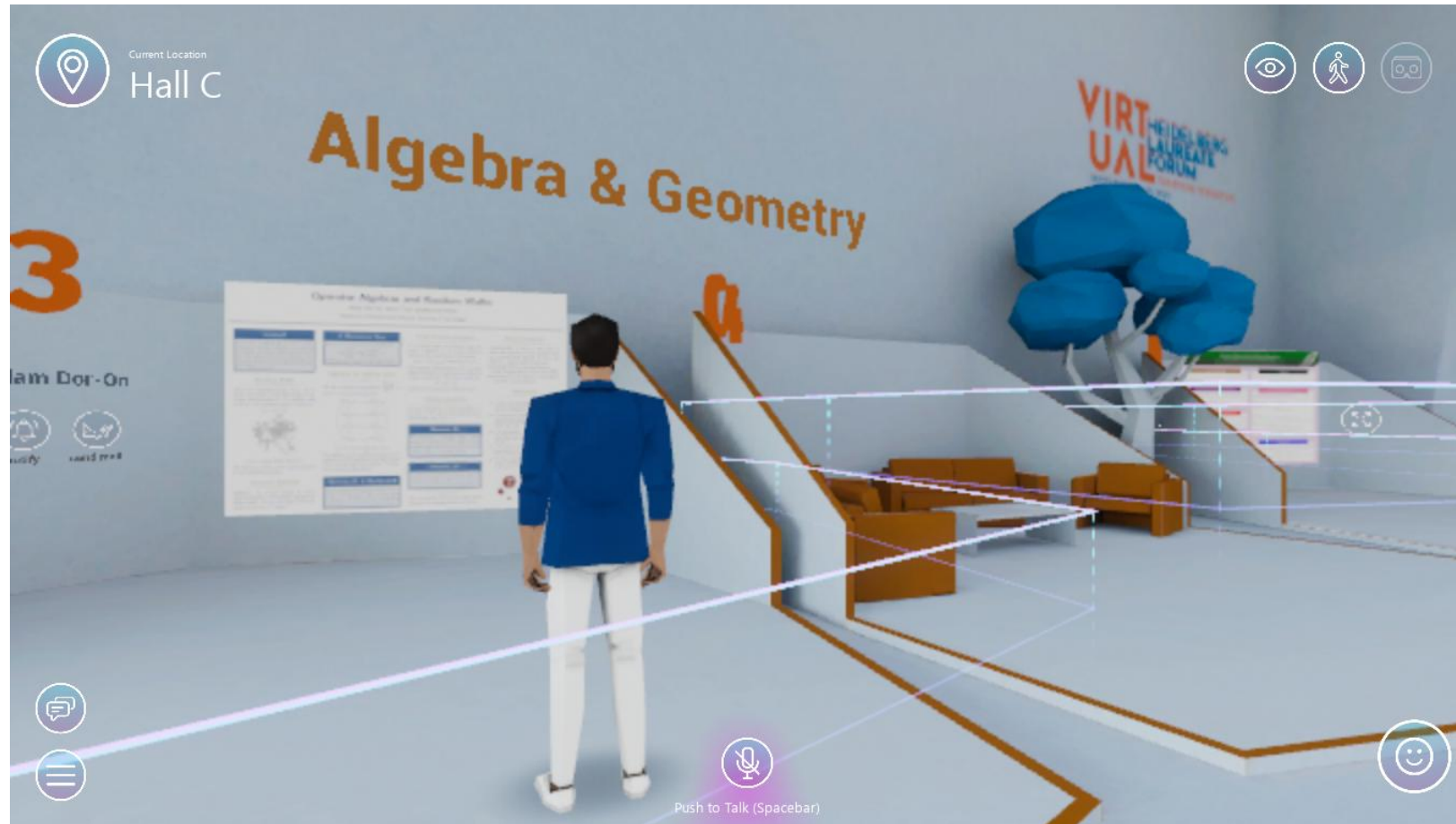
3D Virtual Event: Poster Sessions



3D Virtual Event: Poster Sessions



3D Virtual Event: Poster Sessions

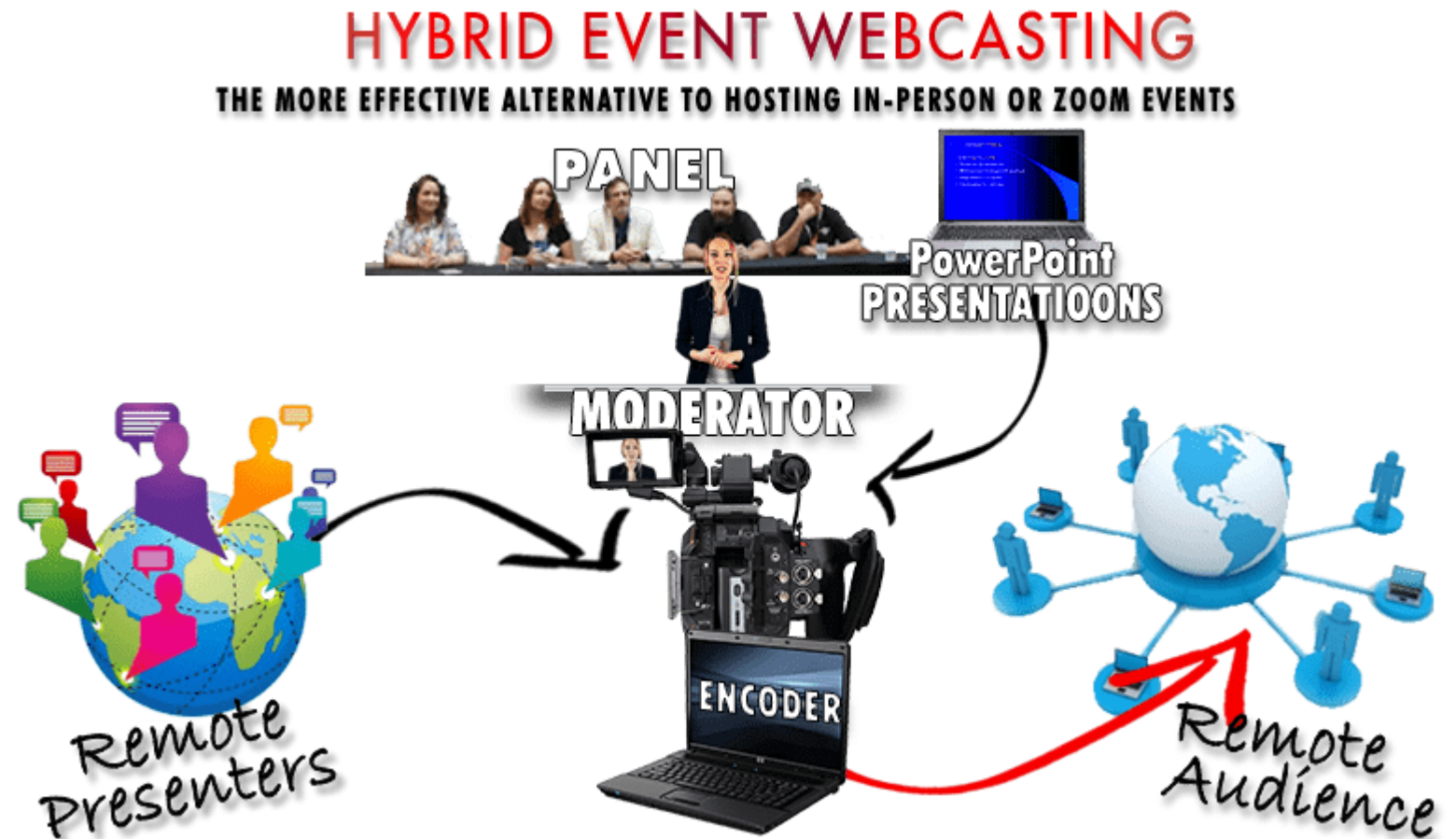


3D Virtual Event: Recreations



Hybrid Event

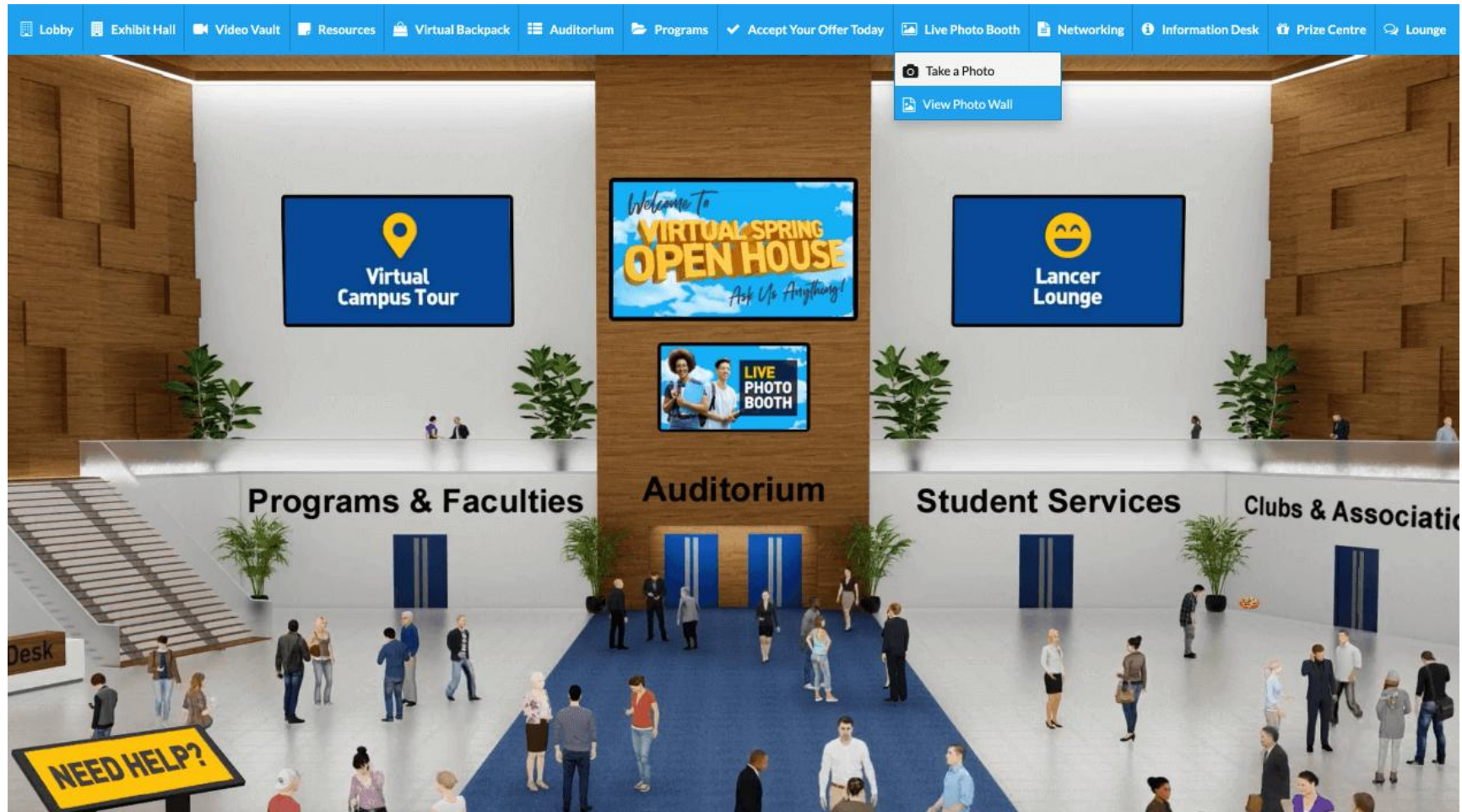
- Combines a “physical” in-person event with a "virtual" online component
- Tradeshow, Conference, Seminar, Workshop, Convocation



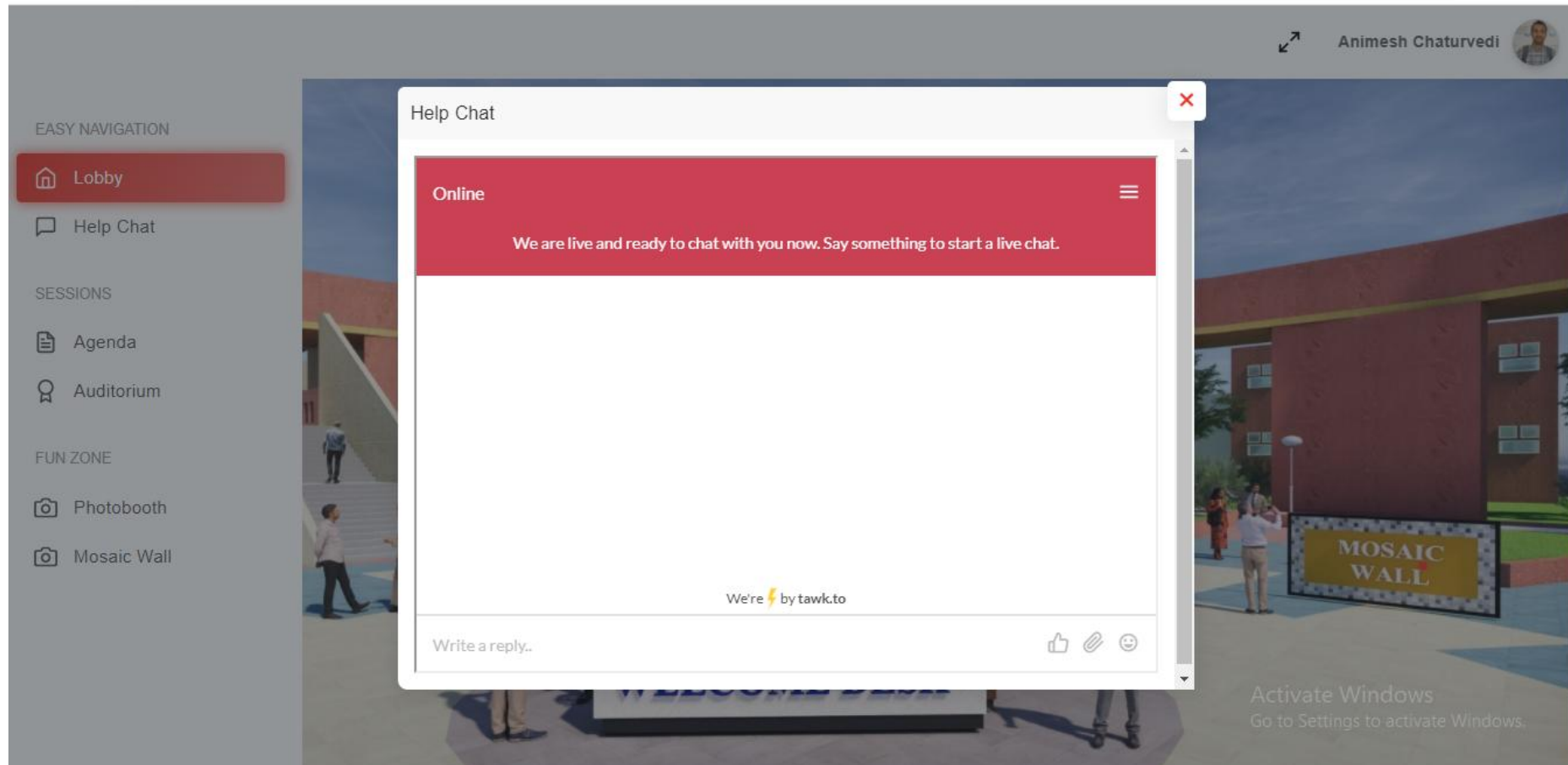
Hybrid Event: Landing Page



Hybrid Event: Landing Page



Hybrid Event: Help Chat FrontEnd



Hybrid Event: Help Chat BackEnd

The screenshot displays a web-based chat interface for a hybrid event. The interface is divided into three main sections: a left sidebar, a central chat area, and a right sidebar.

Left Sidebar: Contains navigation icons for home, settings, user profile, and search. Below these are sections for "Groups" and "Direct Messages", each with a search icon. A list of chat thumbnails is visible at the bottom.

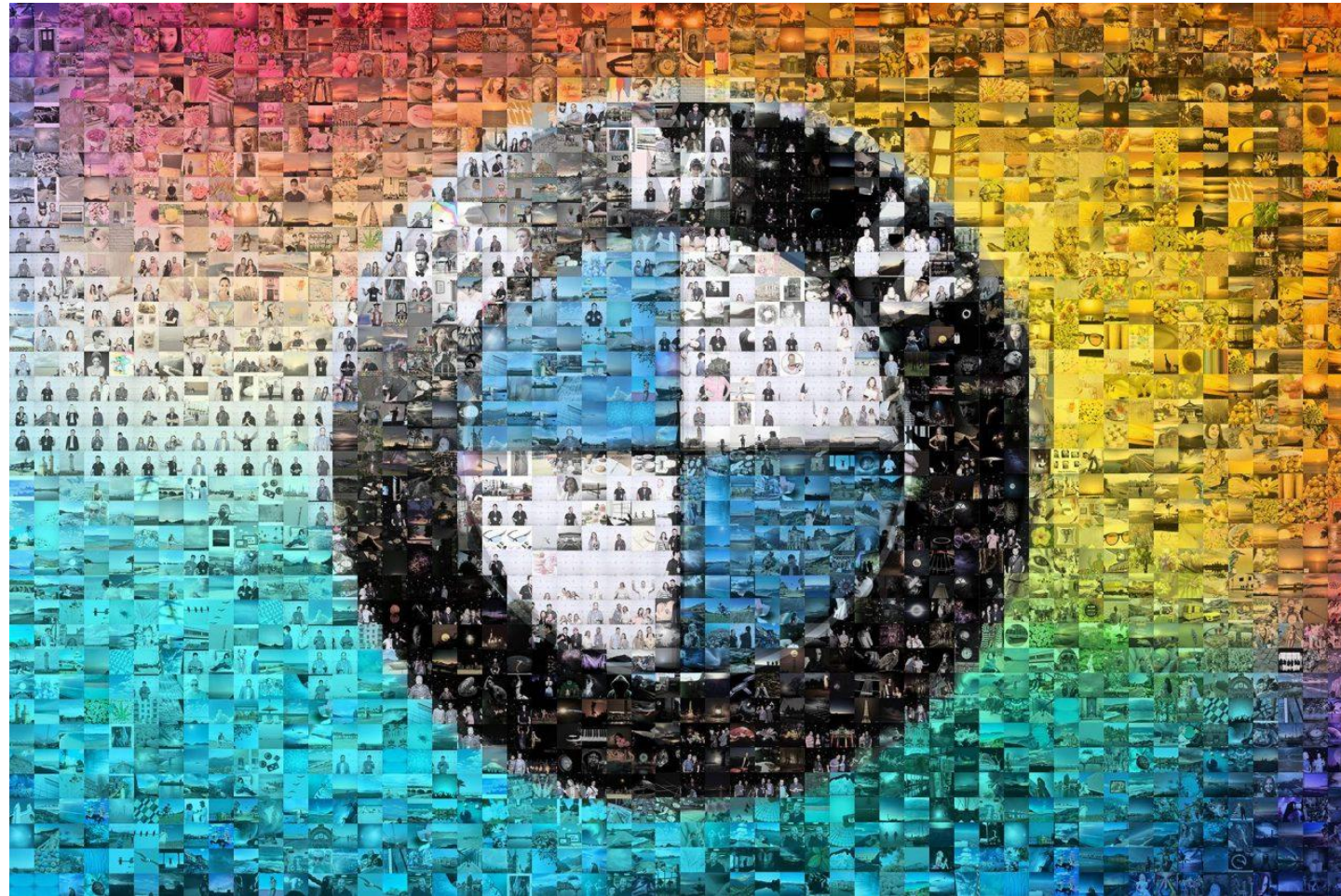
Central Chat Area: Shows a chat history with messages from a visitor (V1605950166179364) and the host. The messages are timestamped 16:25 and 16:27. The chat history includes:

- Visitor: hi (16:25)
- Visitor: hello (16:25)
- Host: [Redacted] (16:25)
- Visitor: hi (16:25)
- Visitor: Amazing work (16:25)
- Visitor: Bye! (16:25)
- System: Visitor closed the popout window (16:25)
- Host: [Redacted] (16:25)
- Visitor: thanks 🙌 (16:25)
- System: Visitor left (16:27)

Right Sidebar: Contains a search bar, a list of chat thumbnails, and a section for "Active Chats". The "Active Chats" section shows a list of chat sessions with their status and duration.

Bottom Bar: Contains a "Message" button and a "Whisper" button.

Hybrid Event: Mosaic Wall



<https://www.youtube.com/watch?v=BSwJCRKvmPc>

Hybrid Event: Blowing Flower Effect

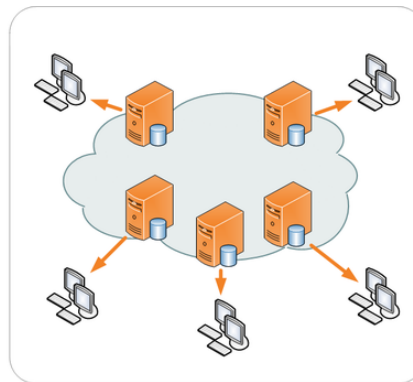
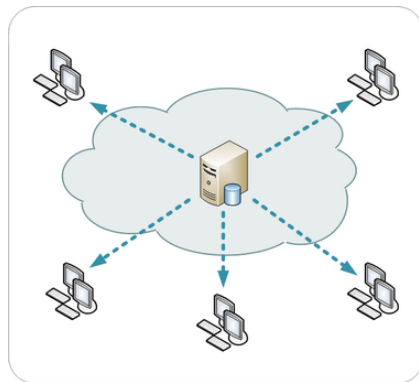


Edge Caching

1. **Content Delivery Network (CDN)**
2. **Facebook and Akamai Data Caching**

Content Delivery Network, or Content Distribution Network (CDN)

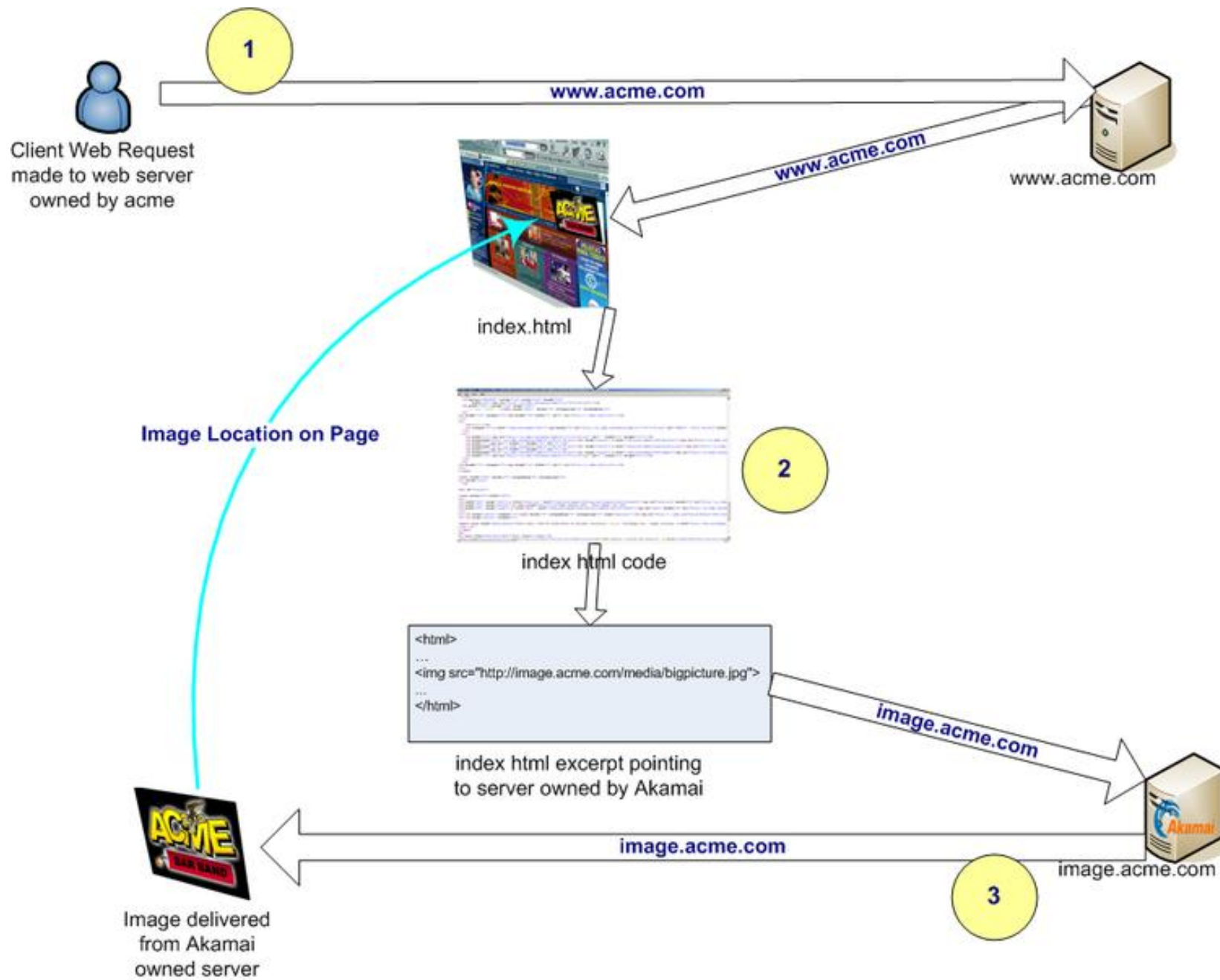
- CDN is a geographically distributed network of proxy servers and their data centers
 - to provide high availability and performance by distributing the service spatially relative to end users.
 - came in late 1990s to improve the performance bottlenecks of the Internet
- Peer-to-peer CDNs: Consumers provide resources as well as use them. Different from client-server
- Private CDNs: Content owners create their own CDN.



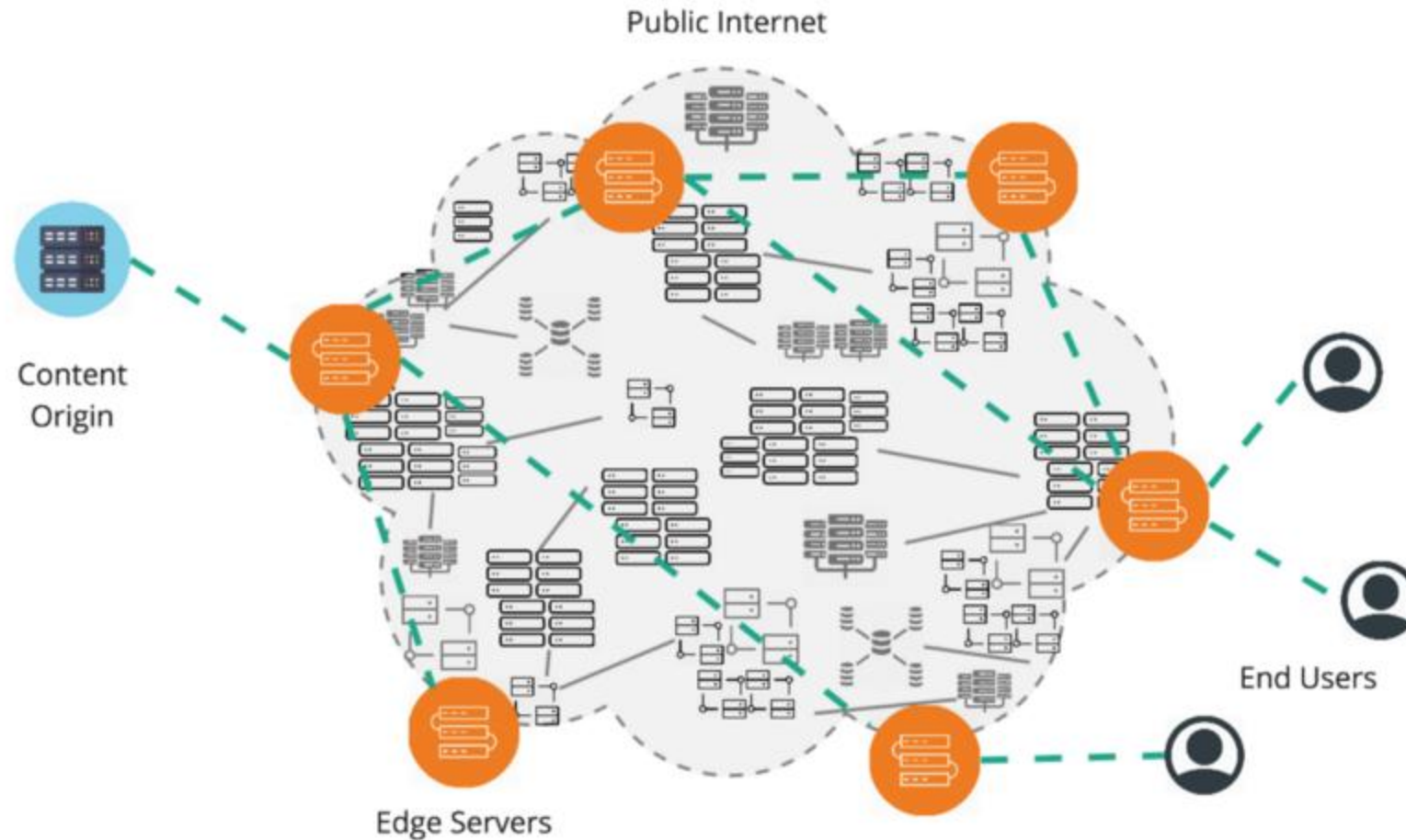
(Left) Single server distribution
(Right) CDN scheme of distribution

Akamai CDN

- Global CDN, Cybersecurity, and Cloud service provider.
- Akamai CDN is one of the largest distributed computing platform
 - Serving between 15% and 30% of all web traffic
 - Operates and rents worldwide network of servers
 - Consumer websites work faster by distributing content from locations near the end-user.
 - End-user navigates to the URL of an Akamai's consumer; their browser is redirected to one of Akamai's copies of the website.
- Facebook as Consumer of Akamai CDN uses many edge servers closer to FB consumer (i.e. FB users).



Akamai Content Delivery Network

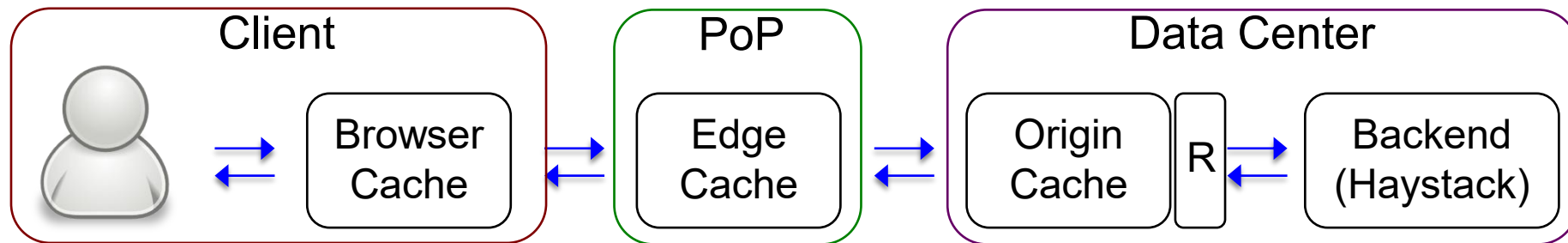


Hybrid Akamai and Facebook

- Facebook Photo Serving Stack: components are linked to show the photo retrieval work-flow.
- Desktop and Mobile clients initiate request traffic, which routes either directly to the Facebook Edge or via Akamai CDN depending on the fetch path.
- The Origin Cache collects traffic from both paths, serving images from its cache and resizing them if needed.
- The Haystack backend holds the actual image content.

Facebook's Photo-Serving Stack

- **Browser:** The first cache layer is in the client's browser
- **Edge:** The Facebook Edge is comprised of a set of Edge Caches that each run inside points of presence (PoPs) close to end users.
- **Haystack:** The backend layer accessed when there is a miss in the Origin cache (co-located with storage servers), the image can often be retrieved from a local Haystack server.

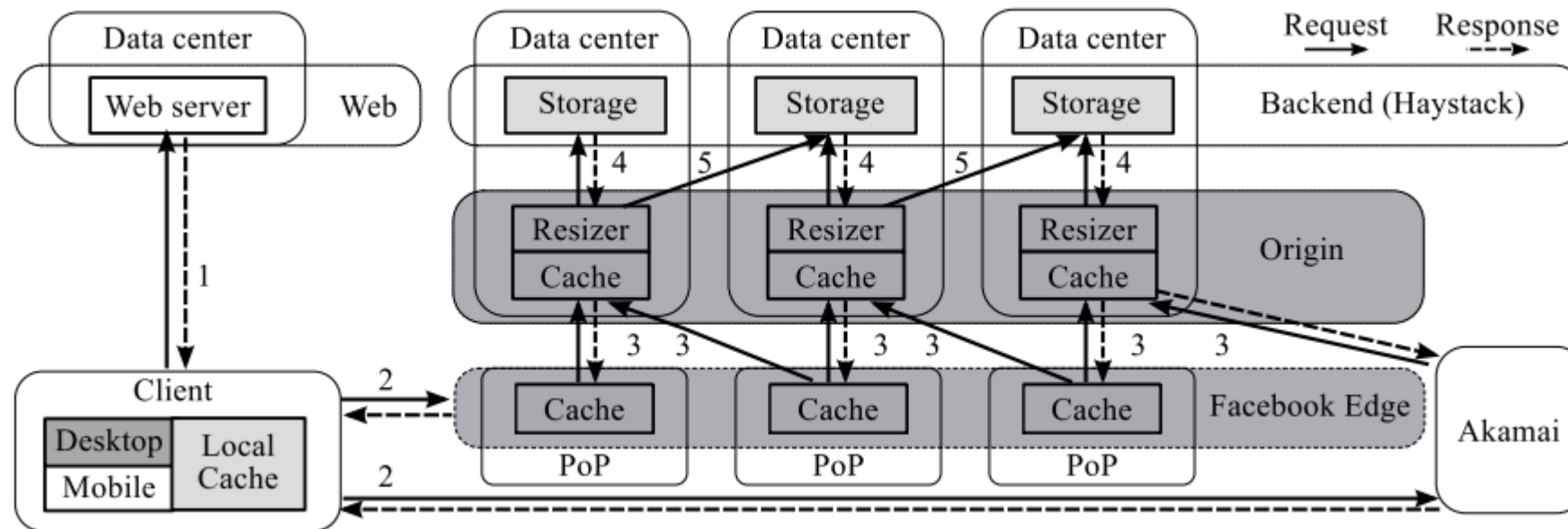


Huang, Qi, et al. "An analysis of Facebook photo caching." *Proceedings of the Twenty-Fourth ACM Symposium on Operating Systems Principles*. 2013.

Hybrid Akamai and Facebook

Architecture and full life cycle of a photo request

- Photo Transformations: Resizers transform photos that are requested by the Akamai CDN,
- Caching Stack



Huang, Qi, et al. "An analysis of Facebook photo caching." *Proceedings of the Twenty-Fourth ACM Symposium on Operating Systems Principles*. 2013.

References

- Weiser, Mark. "The computer for the 21st century." *ACM SIGMOBILE mobile computing and communications review* 3.3 (1999): 3-11.
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- <https://en.wikipedia.org/wiki/Cloudlet>
- IEEE Standard Association. "IEEE 1934-2018-IEEE Standard for adoption of OpenFog reference architecture for fog computing." (2018).
- https://en.wikipedia.org/wiki/Fog_computing
- https://en.wikipedia.org/wiki/Virtual_event
- [Virtual HLF 2020 - Heidelberg Laureate Forum](#)
- https://en.wikipedia.org/wiki/Hybrid_event
- https://en.wikipedia.org/wiki/Akamai_Technologies
- https://en.wikipedia.org/wiki/Content_delivery_network
- Huang, Qi, et al. "An analysis of Facebook photo caching." *Proceedings of the Twenty-Fourth ACM Symposium on Operating Systems Principles*. 2013.

תודה רבה

Hebrew

Ευχαριστώ

Greek

Спасибо

Russian

Danke

German

धन्यवादः

Sanskrit

நன்றி

Tamil

شكراً

Arabic

Merci

French

ধন্যবাদ
Bangla

ಧನ್ಯವಾದಗಳು
Kannada

Thank You

English

നന്നി

Malayalam

Grazie

Italian

ధన్యవాదాలు
Telugu

આભાર
Gujarati

多謝

Traditional Chinese

Gracias

Spanish

ਧੰਨਵਾਦ
Punjabi

धन्यवाद

Hindi & Marathi

多谢

Simplified Chinese

<https://sites.google.com/site/animeshchaturvedi07>

Obrigado

Portuguese

ありがとうございました

Japanese

ขอบคุณ

Thai

감사합니다

Korean